Lab 1: Meet SPSS — Variables, Measurement, and Your First Dataset

Covers chapter 1 material: Introduction to Statistics | 50-minute lab

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Before you start: Open SPSS Statistics on your lab computer. When it opens, you may see a "What's New" or "Welcome" dialog box — close it. You should be looking at a blank **Data Editor** window with a grid of empty cells.

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1 Part 0: Get Oriented (3 min)

Look at the bottom-left corner of the Data Editor window. You'll see two tabs:

- **Data View** — where you type in scores, one row per person, one column per variable
- **Variable View** — where you describe each variable *before* you enter any scores

There is also a separate window called the **Output Viewer** — it doesn't exist yet, but it will pop up automatically the first time you run an analysis. That's where your results/tables appear.

Rule for today: we always set up a variable in **Variable View** *before* typing any scores for it in **Data View**. This mirrors Chapter 1's point that researchers must define how something will be measured before they collect data.

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2 Part 1: Build Your Variable View (17 min)

You're going to enter data from a mini-survey of 10 students. Click the **Variable View** tab now. You'll see rows (one per variable you're about to create) and columns including Name, Type, Decimals, Label, Values, and Measure.

We will create **6 variables**, one per row. For each one, follow these exact steps.

2.1 Row 1: StudentID

1. Click the empty cell in **row 1, "Name" column**. Type: `StudentID`
2. Leave **Type** as the default (Numeric).
3. Click the **Decimals** cell for this row. Change it to `0` (an ID number has no decimal places).
4. Click the **Label** cell. Type: `Student identifier`
5. Click the **Measure** cell. Click the dropdown arrow and select **Nominal**.
   - *Why nominal?* An ID number is just a name written as a digit — Student #7 isn't "more" than Student #3. This is exactly the room-number example from the textbook: room numbers are labels, not quantities.

2.2 Row 2: Major

1. **Name:** `Major`
2. **Decimals:** `0`
3. **Label:** `Academic major`
4. **Values:** Click the Values cell, then click the small gray **...** button that appears on the right side of the cell. A "Value Labels" dialog opens.
   - In the **Value** box, type `1`. In the **Label** box, type `Psychology`. Click **Add**.
   - Value `2`, Label `Biology`. Click **Add**.
   - Value `3`, Label `Business`. Click **Add**.
   - Value `4`, Label `Undecided`. Click **Add**.
   - Click **OK**.
5. **Measure:** select **Nominal**.

2.3 Row 3: ClassYear

1. **Name:** `ClassYear`
2. **Decimals:** `0`
3. **Label:** `Year in school`
4. **Values:** same process as above —
   - `1` = `Freshman`, `2` = `Sophomore`, `3` = `Junior`, `4` = `Senior` (Add after each one, then OK)
5. **Measure:** select **Ordinal**.
   - *Why ordinal, not nominal?* Freshman → Senior is an ordered sequence, unlike majors.

2.4 Row 4: StudyHours

1. **Name:** `StudyHours`
2. **Decimals:** change to `1` (allow one decimal place, e.g., 8.5)
3. **Label:** `Hours spent studying per week`
4. **Values:** leave blank (this variable's numbers are real quantities, not category codes — nothing to label)
5. **Measure:** select **Scale**.

2.5 Row 5: NetflixHours

1. **Name:** `NetflixHrs`
2. **Decimals:** `1`
3. **Label:** `Hours streaming video per week`
4. **Values:** leave blank
5. **Measure:** select **Scale**.

2.6 Row 6: StressLevel

1. **Name:** `Stress`
2. **Decimals:** `0`
3. **Label:** `Self-rated stress, 1 (low) to 10 (high)`
4. **Values:** leave blank for now — we'll come back to this one in the discussion question below
5. **Measure:** select **Scale** for now (we'll revisit whether this is the right call)

> Checkpoint
>
> You should have 6 rows filled in. Raise your hand if any Measure cell still says "Unknown."

> Note on SPSS's Measure column
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> The textbook describes **four** scales of measurement (nominal, ordinal, interval, ratio). SPSS only gives you **three** options. Hang onto that observation — it's Concept Check question 4 below.

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3 Part 2: Enter the Data (15 min)

Click the **Data View** tab. Your 6 column headers (StudentID, Major, ClassYear, StudyHours, NetflixHrs, Stress) should already be there from Part 1. Type the following, one row per student — click a cell, type the number, press **Tab** or **Enter** to move to the next cell.

StudentID	Major	ClassYear	StudyHours	NetflixHrs	Stress
1	1	2	12.5	6.0	7
2	3	4	6.0	10.5	5
3	2	1	15.0	3.5	8

StudentID	Major	ClassYear	StudyHours	NetflixHrs	Stress
4	4	1	8.5	9.0	6
5	1	3	10.0	5.5	4
6	2	4	18.5	2.0	9
7	3	2	7.5	12.0	5
8	1	4	11.0	4.5	6
9	4	3	9.5	8.0	7
10	2	2	13.0	6.5	8

Tip

In Data View, the Major and ClassYear columns will still show plain numbers (1, 2, 3, 4) unless you turn on value labels. To check your coding: go to the **View** menu (top menu bar) → click **Value Labels**. The columns should now display "Psychology," "Sophomore," etc. instead of numbers. Toggle it back off and on to see both views — the number is what's stored, the label is what humans read.

Save your file now: File → Save As → name it `Lab1_YourLastName.sav` → save it to the location your instructor specifies.

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4 Part 3: Run Your First Output (9 min)

We haven't learned to *interpret* statistical output yet — that starts in Chapter 2. Right now we're just previewing what "descriptive statistics" (Chapter 1, Learning Objective 2) looks like in SPSS.

3a. Frequency table for a nominal variable: 1. Menu bar: **Analyze → Descriptive Statistics → Frequencies** 2. In the dialog box, click **Major** in the left-hand list, then click the arrow button to move it into the "Variable(s)" box on the right. 3. Click **OK**. 4. The Output Viewer window opens automatically with a table showing how many students fall into each major category, and what percentage each represents.

3b. Summary numbers for a scale variable: 1. **Analyze → Descriptive Statistics → Descriptives** 2. Move **StudyHours** and **NetflixHrs** into the "Variable(s)" box. 3. Click **OK**. A second table appears in the Output Viewer with N, Minimum, Maximum, Mean, and Std. Deviation for each.

Save your output too: in the Output Viewer window, File → Save As → `Lab1_YourLastName_Output.spv`.

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5 Part 4: In-Lab Concept Checkpoint (6 min — complete before you leave)

Answer these on paper or in the LMS quiz (whichever your instructor specifies):

1. If we wanted to generalize beyond this room, what would the **population** be, and what would the 10 rows of data represent (in Chapter 1's terms)?
2. The average StudyHours you'd calculate from these 10 rows — is that a **parameter** or a **statistic**? Why?
3. When you ran Frequencies and Descriptives in Part 3, were you using **descriptive** or **inferential** statistics? How do you know?

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6 Take-Home Lab Report (due before next class)

Using your saved `.sav` file and output, write brief answers (a sentence or two each is fine):

1. **Variable classification table.** For each of the 6 variables, state (a) whether it's discrete or continuous, and (b) which of the four textbook scales of measurement it belongs to (nominal/ordinal/interval/ratio).
2. **SPSS's "Scale" category.** SPSS collapses two of the textbook's four scales into one "Scale" option. Which two, and why does that combination make statistical sense (hint: think about what interval and ratio scales have in common that nominal and ordinal don't)?
3. **Real limits.** One classmate's StudyHours value is 8.5. What are the real limits of that score, and why do real limits only apply to continuous variables?
4. **Stress variable debate.** We treated Stress (a 1–10 self-rating) as Scale in SPSS, but reasonable people disagree. Make a one-paragraph case that it's actually *ordinal* instead, using the textbook's definition of what an ordinal scale can and can't tell you.
5. **Independent/dependent variables.** Suppose a researcher used this same setup to test whether *Major* is related to *StudyHours* — comparing average study hours across the four majors. Identify the independent variable and the dependent variable. Is the independent variable a true independent variable or a **quasi-independent** variable? Explain using the textbook's distinction.
6. **Data structure.** Which of the three data structures from Section 1-3 (one-group descriptive research, the correlational method, or comparing groups of scores) best describes what we did today with the full 6-variable dataset? Justify your answer.

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7 Instructor Notes

! Trim this section before distributing to students

- **Timing is tight** — Part 1 (variable setup) is where students lose time fumbling with the Values dialog. Consider projecting your own screen through Row 1 and Row 2 live, then letting them do Rows 3–6 independently.
- **Common error:** students forget to click "Add" before "OK" in the Value Labels dialog, so labels don't save. Worth a 10-second warning up front.
- **Common error:** leaving Decimals at the default (2) for StudentID/Major/ClassYear — not wrong mechanically, but a good moment to ask "does a major code need decimal places?"
- Question 4 (Stress variable) has no single correct answer — it's the intro-stats version of the ordinal-vs-interval Likert debate, and is meant to generate discussion, not a graded "right answer." Grade for reasoning quality, not conclusion.
- If a pair finishes early, have them toggle **View → Value Labels** off/on a few times and predict what the Frequencies output would look like using raw codes vs. labels — reinforces that the label is cosmetic, not the stored value.